Notes: Fleiner, Jagadeesan, Janko & Teytelboym (Econometrica, 2019)

Trading Networks with Frictions

Contents

1 Big picture

- **Question.** How do continuous transfers, discrete bilateral trades, and distortionary frictions interact in trading networks?
- **Motivation.** Modern production is a network. Firms buy from suppliers and sell to customers, often through bilateral contracts with non-price terms, delivery obligations, and prices.
- **Main novelty.** The paper allows continuous prices but also imperfect transferability caused by transaction taxes, commissions, or other wedges between what one side pays and the other side effectively receives.
- **Core equilibrium concept.** A competitive equilibrium is an arrangement of trades and prices such that every firm demands exactly its realized trades at those prices.
- **Existence result.** If firms' choices are fully substitutable and compensating variations are bounded, competitive equilibria exist.
- **Welfare warning.** With distortionary frictions, competitive equilibria need not be Pareto-efficient and need not be stable in the usual cooperative sense.
- **Cooperative solution concept.** The paper uses *trail stability*, which rules out sequential blocking deviations along paths in the trading network.
- **Equivalence result.** Competitive equilibrium outcomes are trail-stable. Under full substitutability and bounded willingness to pay, every trail-stable outcome can be supported as a competitive equilibrium.
- **Economic takeaway.** Price-taking equilibrium and a cooperative notion of local network stability are essentially equivalent under substitutability, even with frictions. But this equivalence does not restore Pareto efficiency or core stability.

2 Objects and measurement in the theory

2.1 Firms, trades, contracts, and outcomes

- There is a finite set of firms F and a finite set of trades Ω.
- Each trade $\omega \in \Omega$ has a seller $s(\omega)$ and a buyer $b(\omega)$.
- A contract is a pair
$$x = (\omega, p_\omega),$$
where $p_\omega \in \mathbb{R}$ is the price of trade ω.
- The full contract set is
$$X = \Omega \times \mathbb{R}.$$
- An outcome is a set of contracts with at most one price for each trade.
- An arrangement is a pair $[\Xi; p]$, where $\Xi \subseteq \Omega$ is a set of realized trades and $p \in \mathbb{R}^\Omega$ specifies prices for all trades, including unrealized trades.
- The associated outcome is
$$\kappa([\Xi; p]) = \{(\omega, p_\omega) : \omega \in \Xi\}.$$

Interpretation: The distinction between outcomes and arrangements is important. Outcomes only list realized contracts. Competitive equilibria require prices for unrealized trades too, because firms must not want to add them.

2.2 Upstream and downstream trades

For a firm f:

- $\Omega_{\to f}$ denotes trades in which f is the buyer, or upstream purchases.
- $\Omega_{f\to}$ denotes trades in which f is the seller, or downstream sales.
- $\Omega_f = \Omega_{\to f} \cup \Omega_{f\to}$ denotes all trades involving f.

Interpretation: The upstream/downstream language is what lets the paper model supply chains and general production networks rather than just two-sided markets.

2.3 Utility functions

Each firm has a utility function

$$u^f : \mathcal{P}(\Omega_f) \times \mathbb{R}^{\Omega_f} \to \mathbb{R} \cup \{-\infty\}.$$

The first argument is the set of trades that involve f; the second is the full transfer vector associated with those trades.

The model assumes:

- continuity in transfers;
- monotonicity in transfers;
- autarky is feasible: $u^f(\varnothing, 0) \in \mathbb{R}$;
- $u^f(\Xi, t) = -\infty$ may be used to represent technological infeasibility.

The transferable-utility special case is

$$u^f(\Xi, t) = v^f(\Xi) + \sum_{\omega \in \Omega_f} t_\omega.$$

Interpretation: The paper's key generalization is that utility can depend on every trade-specific transfer separately, not only on the sum of transfers. This is how frictions break perfect transferability.

2.4 Demand correspondence

At a price vector p, firm f receives positive transfers on sales and pays negative transfers on purchases. Its demanded trade sets are

$$D^f(p) = \arg\max_{\Xi \subseteq \Omega_f} u^f\big(\Xi, (p_{f\to}, (-p)_{\to f}, 0_{\Omega_f \setminus \Xi})\big).$$

Interpretation: Demand is over trade bundles, not individual trades. A firm may demand a sale only if it can also obtain a complementary input purchase.

2.5 Choice correspondence from available contracts

Given an available set of contracts $Y \subseteq X_f$, firm f chooses

$$C^f(Y) = \arg\max_{Z \subseteq Y,\ Z \text{ outcome}} U^f(Z),$$

where $U^f(Z)$ is the utility generated by the contracts in Z.

Interpretation: The demand correspondence is used for competitive equilibrium prices. The choice correspondence is used for matching-theoretic stability concepts.

3 Models and theoretical specifications

3.1 Competitive equilibrium

An arrangement $[\Xi; p]$ is a competitive equilibrium if

$$\Xi_f \in D^f(p_{\Omega_f}) \quad \text{for all } f \in F.$$

Interpretation: Markets clear because buyers and sellers demand the same trades. A trade occurs only when both sides demand it, and a trade fails to occur when at least one side does not demand it.

3.2 Transaction taxes and imperfect transferability

For a proportional transaction tax λ, the net-of-tax transfer function is

$$T_\omega(t_\omega) = \begin{cases} (1-\lambda)t_\omega, & t_\omega \geq 0, \\ t_\omega, & t_\omega < 0. \end{cases}$$

If firms are quasilinear before taxes, utility becomes

$$u^f(\Xi, t) = v^f(\Xi) + \sum_{\omega \in \Omega_f} T_\omega(t_\omega).$$

More generally, if $\widehat{u}^f$ is the pre-tax utility function, then

$$u^f(\Xi, t) = \widehat{u}^f\big(\Xi, (T_\omega(t_\omega))_{\omega \in \Omega_f}\big).$$

Interpretation: A tax or commission creates a wedge: the seller values a payment net of tax, while the buyer pays the gross amount. Hence utility is not perfectly transferable even when preferences are otherwise quasilinear.

3.3 Why competitive equilibrium can be inefficient

The paper's leading examples show two failures.

- **Example 1.** In a two-firm trading cycle with a transaction tax, autarky can be a competitive equilibrium even though joint trade creates gains. The tax wedge makes the price system unable to support the efficient trade.
- **Example 2.** Adding an unused outside option can shut down trade between two other firms. The outside option forces a high price on one contract; the transaction tax on that price destroys the surplus from another trade.

Interpretation: In frictionless transferable-utility economies, the First Welfare Theorem rules out such Pareto-inefficient competitive equilibria. With distortionary frictions, the theorem fails.

3.4 Full substitutability

Full substitutability requires three economic properties:

- upstream purchases are substitutes for each other;
- downstream sales are substitutes for each other;
- upstream and downstream trades are complements.

In the paper's choice-language version, when a firm's upstream opportunities expand and downstream opportunities contract, previously chosen downstream contracts remain attractive and upstream contracts become weakly less attractive; the opposite comparative statics apply when downstream opportunities expand and upstream opportunities contract.

Interpretation: Full substitutability allows production-chain complementarity between inputs and outputs, but it rules out problematic complementarities within the set of inputs or within the set of outputs.

3.5 Bounded compensating variations

The bounded compensating variations condition is

$$\inf_{(\Xi,t):u^f(\Xi,t)>u^f(\varnothing,0)} \sum_{\omega\in\Omega_f} t_\omega > -\infty.$$

Interpretation: No bundle of trades is so desirable that a firm would accept arbitrarily negative net transfers to obtain it. This is a mild real-world regularity condition but is necessary for existence with unbounded prices.

3.6 Existence of competitive equilibrium

Theorem 1. Under full substitutability and bounded compensating variations, competitive equilibria exist.

The proof has three steps:

1. modify preferences by giving firms expensive options to execute trades, so willingness to pay is bounded;
2. use bounded compensating variations to show that competitive equilibria of the modified economy do not use those artificial options;
3. discretize prices, use a generalized deferred-acceptance operator, and take limits.

Interpretation: The theorem extends existence results for transferable-utility trading networks to environments with taxes, commissions, and imperfect transferability.

3.7 Individual rationality and stability

An outcome A is individually rational if

$$A_f \in C^f(A_f) \quad \text{for all } f.$$

Stability rules out group deviations: no nonempty set of new contracts $Z \subseteq X \setminus A$ can block the outcome while firms are allowed to keep or drop existing contracts.

Interpretation: Stability is too demanding in general trading networks with frictions. Competitive equilibrium outcomes can be unstable, and stable outcomes need not exist.

3.8 Trail stability

A trail is a sequence of contracts $(x_1, \ldots, x_n)$ such that

$$b(x_i) = s(x_{i+1}) \quad \text{for } 1 \leq i < n.$$

A trail locally blocks an outcome if the first seller wants to offer the first contract, each intermediate firm can accept the incoming offer and make the next outgoing offer, and the last buyer can accept the final contract.

An outcome is trail-stable if it is individually rational and has no locally blocking trail.

Interpretation: Trail stability is a sequential, local deviation concept. It captures the idea that contracts are proposed along supply chains or trading paths rather than by arbitrary coalitions that coordinate all at once.

3.9 Competitive equilibrium implies trail stability

Theorem 2. Every competitive equilibrium outcome is trail-stable.

Interpretation: At equilibrium prices, the first seller in a blocking trail would require a price above the equilibrium price, while the last buyer would require a price below the equilibrium price. Along the trail these inequalities cannot all hold simultaneously.

3.10 Bounded willingness to pay and lifting trail stability

Bounded willingness to pay requires the existence of M such that no firm is willing to buy any trade at price at least M, and no firm is willing to sell any trade at price at most $-M$.

Theorem 3. Under full substitutability and bounded willingness to pay, every trail-stable outcome lifts to a competitive equilibrium.

Corollary 2. Under full substitutability and bounded willingness to pay, competitive equilibrium outcomes and trail-stable outcomes exist and coincide.

Interpretation: Trail stability is not just an ad hoc cooperative refinement. Under the regularity condition, it has exact competitive foundations.

4 Identification and proof logic

4.1 What drives the results

- **Substitutability** controls strategic complementarity enough for lattice and deferred-acceptance methods to work.
- **Continuous prices** allow competitive prices to support trades, but prices must be assigned even to unrealized trades.
- **Frictions** break perfect transferability, so competitive equilibrium is not necessarily Pareto-efficient or core-stable.
- **Trail deviations** are weak enough to exist in general networks but strong enough to match competitive equilibrium predictions.

Interpretation: The paper's conceptual contribution is not a new welfare theorem. It is a precise equivalence between a price-taking notion and a sequential cooperative notion in networks with frictions.

4.2 Why standard stability fails

- A stable block lets a group of firms coordinate simultaneous recontracting.
- In networks with frictions, such coordinated blocks can exist even when the current outcome is competitive.
- Stable outcomes can fail to exist because different potential blocks chase each other around the network.
- Trail stability avoids this by considering sequential local offers rather than arbitrary group commitments.

Interpretation: The correct cooperative analogue of competitive equilibrium in a network with frictions is trail stability, not core stability or standard stability.

4.3 What the paper cannot claim

- The paper gives existence and equivalence results, not an empirical test.
- Frictions are modeled in reduced form; the paper does not estimate transaction-tax or commission wedges.
- Full substitutability rules out some real production complementarities, especially complementarities among inputs.
- Pareto efficiency is not guaranteed, so equilibrium existence should not be confused with welfare optimality.

Interpretation: The theory is strongest as a benchmark for markets where trades are bilateral, prices are continuous, and local network deviations are economically realistic.

5 Tables and figures

The paper has no empirical tables. The original visual material consists of three figures; because the paper is theoretical, I also include directly cropped equation/result panels from the paper. Adjacent visuals are grouped to save space.

5.1 Model primitives and tax frictions

Read:

- The first panel contains the arrangement-to-outcome map, the utility specification, the transferable-utility special case, and the demand correspondence.
- The second panel contains the transaction-tax transfer function and the net-of-tax utility representation.
- Together, these panels show how the model combines matching with contracts and imperfect transferability.

Economic message: The same object—a bilateral contract with a price—is analyzed both through price-taking demand and through cooperative choice. Frictions enter through the trade-specific transfer function.

a buyer or as a seller). For a set $Y \subseteq X$ of contracts, we define subsets $Y_{\to f}$, $Y_{f\to}$, and Y_f analogously.

An *arrangement* is a pair $[\Xi; p]$ of a set of trades $\Xi \subseteq \Omega$ and a price vector $p \in \mathbb{R}^{\Omega}$. Given an arrangement $[\Xi; p]$, we define an associated outcome $\kappa([\Xi; p]) \subseteq X$ by

$$\kappa([\Xi; p]) = \{(\omega, p_\omega) \mid \omega \in \Xi\}.$$

That is, $\kappa([\Xi; p])$ is the outcome at which the trades in Ξ are realized at the prices given by p. Unlike outcomes, arrangements specify prices even for unrealized trades.

2.2. *Utility Functions and Transfers*

Each firm's utility depends only on the trades that involve the firm and on the transfers that it receives. Formally, firm f has a utility function $u^f : \mathcal{P}(\Omega_f) \times \mathbb{R}^{\Omega_f} \to \mathbb{R} \cup \{-\infty\}$.[3]

Three features of our specification of utility functions are worth highlighting. First, the utility function depends on the entire vector of transfers (as opposed to merely on net transfers). Hence, we allow firms to place different marginal values on transfers associated to different trades; as we show in Section 3, this feature allows our model to capture distortionary frictions. Second, we allow the utility function to take value $-\infty$ to capture technological constraints (Hatfield et al. (2013)). Specifically, we set $u^f(\Xi, t) = -\infty$ for all transfer vectors t if the set Ξ of trades is technologically infeasible for f. Last, while utility depends on the transfers associated to unrealized trades, firms do not receive transfers for unrealized trades in market outcomes.

We assume that $u^f(\Xi, t)$ is continuous in t and that

$$t \le t' \quad \Longrightarrow \quad u^f(\Xi, t) \le u^f(\Xi, t')$$

with equality if and only if $u^f(\Xi, t) = -\infty$, so transfers are relevant to firms whenever a set of trades is feasible. We also assume that $u^f(\emptyset, 0) \in \mathbb{R}$, so autarky is feasible. The transferable utility trading network model of Hatfield et al. (2013) is recovered when

$$u^f(\Xi, t) = v^f(\Xi) + \sum_{\omega \in \Omega_f} t_\omega \tag{1}$$

for some *valuation* function $v^f : \mathcal{P}(\Omega_f) \to \mathbb{R} \cup \{-\infty\}$.

To analyze competitive equilibria, we need to consider firms' demands for trades at any given price vector. Prices give rise to transfers in the following manner. Firms receive no transfer for a trade if they do not agree to the trade. Firms receive transfers equal to the prices of any realized sales (downstream trades) and pay transfers equal to the prices of any realized purchases (upstream trades). Maximizing utility at a price vector $p \in \mathbb{R}^{\Omega_f}$ gives rise to a collection of sets of demanded trades

$$D^f(p) = \underset{\Xi \subseteq \Omega_f}{\arg\max}\, u^f\big(\Xi, (p_{\Xi_{f\to}}, (-p)_{\Xi_{\to f}}, 0_{\Omega_f \setminus \Xi})\big).$$

(a) Model primitives, utility, and demand.

assumption is without loss of generality. Thus, the net-of-tax transfer received or paid by a firm for a trade ω is

$$T_\omega(t_\omega) = \begin{cases} (1-\lambda)t_\omega & \text{if } t_\omega \ge 0, \\ t_\omega & \text{if } t_\omega < 0, \end{cases} \tag{2}$$

where t_ω is the gross transfer (Dupuy et al. (2017)). Hence, when $t_\omega \ge 0$, the firm is a recipient of the transfer and receives $(1-\lambda)t_\omega$; when $t_\omega < 0$, the firm is a payer and pays $|t_\omega|$ in full. As a result, if firm f has quasilinear preferences and valuation function $v^f : \mathcal{P}(\Omega_f) \to \mathbb{R} \cup \{-\infty\}$, then the utility function u^f is

$$u^f(\Xi, t) = v^f(\Xi) + \sum_{\omega \in \Omega_f} T_\omega(t_\omega). \tag{3}$$

When $\lambda < 1$ and $v^f(\emptyset) \in \mathbb{R}$, the utility function u^f satisfies our conditions on preferences (i.e., it is continuous and satisfies the requisite monotonicity condition). Note that transaction taxes make utility imperfectly transferable even if preferences are quasilinear.[7]

We can model transaction taxes similarly even in the presence of income effects. If firm f has utility function $\widehat{u}^f$ before taxes, then the net-of-tax utility function is

$$u^f(\Xi, t) = \widehat{u}^f\big(\Xi, (T_\omega(t_\omega))_{\omega \in \Omega_f}\big). \tag{4}$$

More generally, our framework can capture nonlinear transaction taxes and subsidies. If a tax of $\Lambda_\omega(t_\omega)$ must be paid on a transfer of size $t_\omega \ge 0$ for trade ω, then we can take the net-of-tax transfer function T_ω to be

$$T_\omega(t_\omega) = \begin{cases} t_\omega - \Lambda_\omega(t_\omega) & \text{if } t_\omega \ge 0, \\ t_\omega & \text{if } t_\omega < 0, \end{cases}$$

in (4) to define the net-of-tax utility function. The case of $\Lambda_\omega(t_\omega) = \lambda t_\omega$ recovers the proportional transaction tax discussed above. If $\widehat{u}^f$ is continuous and satisfies the requisite

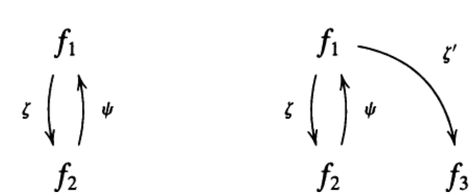
(b) Transaction-tax equations.

5.2 Figure 1 and Figure 2: leading network examples

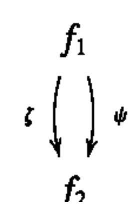
Read:

- Figure 1(a) is a two-firm cyclic economy with two trades. A proportional tax can make autarky a competitive equilibrium even when joint trade creates surplus.
- Figure 1(b) adds a third firm and an outside option. That option is not realized in equilibrium but changes price constraints enough to shut down the original market.
- Figure 2 shows a two-firm network used to demonstrate that trail-stable outcomes need not lift to competitive equilibria when full substitutability fails.

Economic message: The examples show why frictions are not innocuous. They can generate Pareto inefficiency, destroy standard stability, and make the relationship between cooperative and competitive concepts depend on substitutability.



(a) Figure 1: trades in Examples 1 and 2.



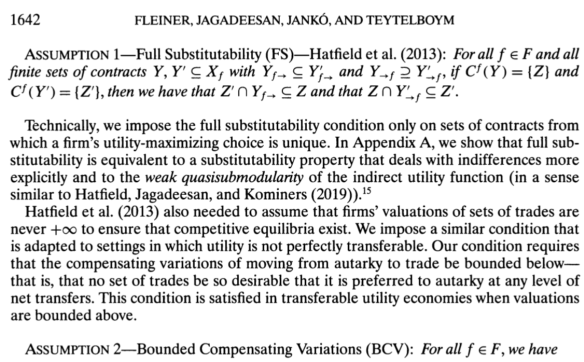
(b) Figure 2: trading in Example 6.

5.3 Assumptions and existence theorem

Read:

- Full substitutability is the main structural condition on firms' contract choices.
- Bounded compensating variations prevents agents from having infinite willingness to accept arbitrarily bad transfers for desirable trades.
- Theorem 1 gives the main existence result: under FS and BCV, competitive equilibria exist.

Economic message: The existence proof combines matching-theoretic substitutability with regularity on compensating variations. This lets the authors handle continuous prices and imperfect transferability simultaneously.



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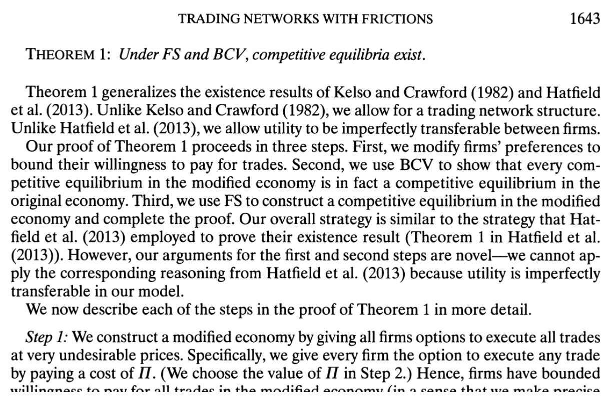
ASSUMPTION 1—Full Substitutability (FS)—Hatfield et al. (2013): *For all $f \in F$ and all finite sets of contracts $Y, Y' \subseteq X_f$ with $Y_{f\to} \subseteq Y'_{f\to}$ and $Y_{\to f} \supseteq Y'_{\to f}$, if $C^f(Y) = \{Z\}$ and $C^f(Y') = \{Z'\}$, then we have that $Z' \cap Y_{f\to} \subseteq Z$ and that $Z \cap Y'_{\to f} \subseteq Z'$.*

Technically, we impose the full substitutability condition only on sets of contracts from which a firm's utility-maximizing choice is unique. In Appendix A, we show that full substitutability is equivalent to a substitutability property that deals with indifferences more explicitly and to the *weak quasisubmodularity* of the indirect utility function (in a sense similar to Hatfield, Jagadeesan, and Kominers (2019)).[15]

Hatfield et al. (2013) also needed to assume that firms' valuations of sets of trades are never $+\infty$ to ensure that competitive equilibria exist. We impose a similar condition that is adapted to settings in which utility is not perfectly transferable. Our condition requires that the compensating variations of moving from autarky to trade be bounded below—that is, that no set of trades be so desirable that it is preferred to autarky at any level of net transfers. This condition is satisfied in transferable utility economies when valuations are bounded above.

ASSUMPTION 2—Bounded Compensating Variations (BCV): *For all $f \in F$, we have*

(a) FS and BCV assumptions.



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THEOREM 1: *Under FS and BCV, competitive equilibria exist.*

Theorem 1 generalizes the existence results of Kelso and Crawford (1982) and Hatfield et al. (2013). Unlike Kelso and Crawford (1982), we allow for a trading network structure. Unlike Hatfield et al. (2013), we allow utility to be imperfectly transferable between firms.

Our proof of Theorem 1 proceeds in three steps. First, we modify firms' preferences to bound their willingness to pay for trades. Second, we use BCV to show that every competitive equilibrium in the modified economy is in fact a competitive equilibrium in the original economy. Third, we use FS to construct a competitive equilibrium in the modified economy and complete the proof. Our overall strategy is similar to the strategy that Hatfield et al. (2013) employed to prove their existence result (Theorem 1 in Hatfield et al. (2013)). However, our arguments for the first and second steps are novel—we cannot apply the corresponding reasoning from Hatfield et al. (2013) because utility is imperfectly transferable in our model.

We now describe each of the steps in the proof of Theorem 1 in more detail.

Step 1: We construct a modified economy by giving all firms options to execute all trades at very undesirable prices. Specifically, we give every firm the option to execute any trade by paying a cost of Π. (We choose the value of Π in Step 2.) Hence, firms have bounded

(b) Theorem 1 and proof roadmap.

5.4 Trail stability and lifting to competitive equilibrium

Read:

- The trail-stability panel defines trails and locally blocking trails.
- The BWP panel states bounded willingness to pay and Theorem 3.
- Theorem 2 says every competitive equilibrium outcome is trail-stable; Theorem 3 gives the reverse direction under FS and BWP.

Economic message: Trail stability is the cooperative solution concept that survives in trading networks with frictions. It is weak enough to exist but strong enough to coincide with competitive equilibrium under regularity.

5.2. Trail Stability

Trail stability is an extension of pairwise stability (in the sense of Gale and Shapley (1962)) to trading networks (Fleiner et al. (2018)). A trail is a sequence of contracts such that the buyer of each contract in the sequence (except for the last) is the seller of the next contract. A trail may involve a firm more than once and can begin and end with contracts that involve the same firm.

DEFINITION 4: A sequence of contracts $(x_1, \dots, x_n)$ is a *trail* if $\mathsf{b}(x_i) = \mathsf{s}(x_{i+1})$ for all $1 \le i \le n-1$.

Trail-stable outcomes are immune to sequential deviations called locally blocking trails. A locally blocking trail begins with a firm f_1 offering a sale contract z_1 that it wishes to sign given its existing contracts, possibly while dropping some existing contracts. The buyer f_2 may accept the offered contract z_1 while dropping some of its existing contracts, in which case a locally blocking trail is formed. The buyer may also hold the proposal z_1 and offer an additional sale contract z_2 to the original proposer or to another firm. This trail of linked offers $z_1, \dots, z_n$ continues until a firm f_{n+1} accepts an offered contract z_n without offering another sale contract, in which case a locally blocking trail is formed.[18]

Our formal definition of trail stability extends the definition given by Fleiner et al. (2018) to settings with indifferences.

DEFINITION 5: A trail $(z_1, \dots, z_n) \in (X \setminus A)^n$ *locally blocks* an outcome A if:
- $z_1 \in Y \setminus A$ for all $Y \in C^{f_1}(A_{f_1} \cup \{z_1\})$, where $f_1 = \mathsf{s}(z_1)$;
- for $1 \le i \le n-1$, we have that $\{z_i, z_{i+1}\} \subseteq Y \setminus A$ for all $Y \in C^{f_{i+1}}(A_{f_{i+1}} \cup \{z_i, z_{i+1}\})$, where $f_{i+1} = \mathsf{b}(z_i) = \mathsf{s}(z_{i+1})$; and
- $z_n \in Y \setminus A$ for all $Y \in C^{f_{n+1}}(A_{f_{n+1}} \cup \{z_n\})$, where $f_{n+1} = \mathsf{b}(z_n)$.

(a) Definition of trail stability.

ASSUMPTION 3—Bounded Willingness to Pay—BWP: *There exists M such that for all firms $f \in F$ and all finite sets of contracts $Y, Z \subseteq X_f$ with $Z \in C^f(Y)$:*
- *If $(\omega, p_\omega) \in Z_{\to f}$, then $p_\omega < M$.*
- *If $(\omega, p_\omega) \in Z_{f \to}$, then $p_\omega > -M$.*

BWP requires that no firm be willing to pay M or more for any trade—that is, that no firm be willing to buy any trade at a price M or more or sell any trade at a price $-M$ or less. Note that BWP rules out many forms of technological constraints, including ones that are permitted under BCV and by Hatfield et al. (2013). In particular, BWP does not allow a firm to require a particular input in order to produce an output, as such constraints would make a firm willing to pay arbitrarily high prices for the input if the firm were able to procure arbitrarily high prices for the output. However, BWP allows for capacity constraints, as they never make trades desirable at extremely unfavorable prices.

BWP helps ensure that trail-stable outcomes lift to competitive equilibria.[21]

THEOREM 3: *Under FS and BWP, trail-stable outcomes lift to competitive equilibria.*

(b) BWP and Theorem 3.

5.5 Figure 3: summary of the paper's results

Read:

- Squiggly arrows represent existence results.
- Solid arrows represent relationships between solution concepts.
- Dashed arrows represent lifting results, where a cooperative outcome can be supported by competitive prices.
- Under FS and BWP, competitive equilibrium and trail stability coincide.
- With no frictions, stronger solution concepts can connect more directly to the core.

Economic message: The figure summarizes the paper's main message: in networks with frictions, trail stability is the right cooperative counterpart to competitive equilibrium, while stability, strong group stability, and the core are generally too strong.

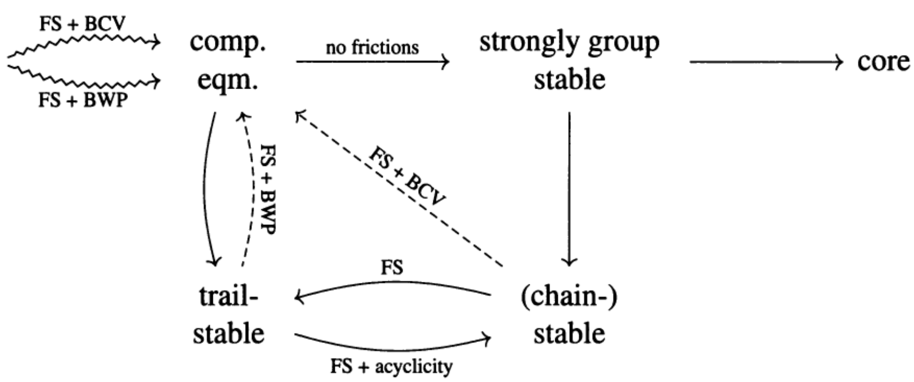

Figure 5: Figure 3: summary of solution-concept relationships.

6 Short summary

- The paper studies trading networks with discrete bilateral trades, continuous prices, and distortionary frictions.
- A contract specifies a trade and a price; an arrangement specifies realized trades and prices for all trades.
- Utility depends on trade bundles and trade-specific transfer vectors, allowing taxes and commissions to make utility imperfectly transferable.
- Competitive equilibrium requires every firm to demand its realized trades at the equilibrium price vector.
- Transaction taxes can make competitive equilibria Pareto-inefficient.
- An outside option that is not used in equilibrium can still shut down another market by altering price constraints.
- Full substitutability means same-side trades are substitutes and cross-side trades are complements.
- Bounded compensating variations rules out infinite willingness to accept bad net transfers for desirable trades.
- Under FS and BCV, competitive equilibria exist.
- Standard stability is too strong: competitive equilibria can be unstable, and stable outcomes may fail to exist.
- Trail stability rules out sequential local deviations along trading trails.
- Every competitive equilibrium outcome is trail-stable.
- Under FS and BWP, every trail-stable outcome can be supported by competitive equilibrium prices.
- Therefore, under regularity, competitive equilibrium and trail stability coincide.
- The equivalence is not a welfare theorem: frictions can still produce inefficient outcomes.

7 Conclusion

7.1 Core conclusion

Fleiner, Jagadeesan, Janko, and Teytelboym show that competitive equilibrium can be studied in trading networks with both continuous prices and distortionary frictions. Under full substitutability and bounded compensating variations, competitive equilibria exist. Under a stronger bounded-willingness-to-pay condition, competitive equilibrium outcomes coincide with trail-stable outcomes.

7.2 Economic intuition

The intuition is that substitutability prevents local contract choices from creating unmanageable demand cycles, while trail stability captures feasible sequential recontracting along the network. Frictions change welfare and standard stability properties, but they do not destroy the connection between prices and local network stability.

7.3 Takeaway

The paper's main lesson is that in markets with supply-chain structure and transaction frictions, the right cooperative benchmark is not the core or standard stability. It is trail stability. Competitive equilibrium remains a powerful concept, but with frictions it should be interpreted as a price-supported trail-stable outcome, not necessarily as an efficient allocation.